324631 (24)

BE (6th Semester)
Examination, April - May, 2016

(Old Scheme)

Fibre Optics

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : Part (*a*) of each question is compulsory carrying 2 marks and solve any two parts from (*b*), (*c*) and (*d*) carrying 7 marks each.

1. (*a*) What are the importances of optical fibre?

(*b*) Prove that for an optical fibre

$$NA = n_1\sqrt{2\Delta}$$

where NA = Numerical aperture

n_1 = Refractive index of core material

Δ = Relative refractive index difference

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(*c*) What is optical fibre? What are the different types of optical fibre and how does light travel in it? Explain with a neat diagram.

(*d*) Describe briefly the salient features of Vapour Axial Deposition (VAD) process and double-crucible method for fibre drawing process.

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2. (*a*) Explain band gap in semiconductor.

(*b*) The quantum efficiency of a particular silicon RAPD is 80% for the detectioin of radiation of wavelength of 0.9 μm, when the incident optical power is 0.5 μW. The output current from the device (after avalanche gain) is 11 μA. Determine the multiplication factor of photo diode under these conditions.

(*c*) Explain Noises in optical communication with their cause.

(*d*) Write short notes on :

(*i*) Optical splices and connectors

(*ii*) Population inversion

3. (*a*) Explain the term 'chromatic dispersion'.

(*b*) Explain signal distortion is waveguide under intramodal dispersion.

(*c*) What do you mean by Attenuation? How does it appear in optical communication?

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(*d*) Explain the term mode coupling. Also explain intermodal dispersion.

4. (*a*) What do you mean by coherent optical detection?

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(*b*) Write a paragraph on optical fiber link design that includes splicing techniques and connection.

(*c*) Explain the modulation techniques for homodyne and heterodyne.

(*d*) Explain link power budget and rise time budget.

5. (*a*) What is wavelength division multiplexing?

(*b*) What are optical amplifiers? Explain.

(*c*) What is FDDI? Briefly discuss FDDI topology.

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(*d*) With the aid of block diagram, describe Synchronous Optical Network (SONET).

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